SWA-MPPI: Socially-Weighted Alignment with Model Predictive Path Integral Control for Cross-Cultural Moral Alignment of Large Language Models

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Abstract

Large language models (LLMs) encode a culturally monolithic moral stance that systematically diverges from the diverse ethical preferences of populations worldwide. We introduce **SWA-MPPI** (**S**ocially-**W**eighted **A**lignment with **M**odel **P**redictive **P**ath **I**ntegral control), a training-free, inference-time framework that steers LLM moral decisions toward country-specific human preferences without modifying model weights. For each target country, SWA-MPPI constructs a small ensemble of culturally grounded *persona agents* from World Values Survey (WVS) Wave 7 data, evaluates their logit-space consensus in a single batched forward pass, and applies a Prospect-Theory-weighted MPPI optimisation step whenever inter-agent disagreement exceeds an adaptively calibrated threshold τ. Evaluated on the Multilingual Trolley Problem (MultiTP) benchmark across 15 countries, 12 native languages, and 6 moral dimensions, SWA-MPPI reduces Jensen-Shannon Distance to human moral preferences by **37.6%** (Qwen2.5-72B) and **34.7%** (Llama-3.1-70B) on average relative to vanilla baselines, while improving Pearson correlation with human preference vectors by up to $+0.41$ points. The framework requires no labelled preference data, operates on frozen weights, and introduces a mean per-scenario overhead of approximately 730 ms on a single NVIDIA H100 GPU.

1 Introduction

As large language models are deployed across increasingly diverse cultural contexts, a fundamental tension has emerged between the uniformity of a single trained model and the plurality of human moral values worldwide. Models trained predominantly on English-language corpora tend to inherit the ethical intuitions of Western, educated, industrialised, rich, and democratic (WEIRD) populations [Henrich et al., 2010], yet they are increasingly enlisted to assist with consequential decisions in societies whose moral norms differ substantially [Awad et al., 2018, Jin et al., 2025].

The Moral Machine experiment [Awad et al., 2018] collated over 40 million human judgments from 233 countries on trolley-problem scenarios, demonstrating profound cross-cultural variation in moral priorities: East Asian respondents weight youth more heavily; Southern-hemisphere populations show distinct utilitarian calculi; Islamic societies prioritise different social roles. The multilingual successor benchmark, MultiTP [Jin et al., 2025], operationalises this evidence into country-specific Average Marginal Component Effects (AMCE) across six moral dimensions and 107 languages, providing a large-scale, cross-culturally validated ground truth for evaluating LLM moral alignment. Against this benchmark, Jin et al. [2025] found that none of 19 evaluated contemporary LLMs achieved broadly satisfactory cross-cultural alignment: most exhibit WEIRD-biased moral reasoning that departs markedly from the graded distributions observed in non-WEIRD human respondents.

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Existing approaches to value alignment fall into two broad families. *Fine-tuning* on human preference data [Bai et al., 2022, Ouyang et al., 2022] can improve cultural fit but demands large per-culture labelled datasets, modifies weights irreversibly, and scales poorly across the 130+ countries in MultiTP. *Prompting-based steering* is lightweight but typically relies on ad-hoc system instructions with no empirical grounding in population-level values, and its effects on cross-lingual moral reasoning are inconsistent [Jin et al., 2025]. Activation-level interventions [Arditi et al., 2025, Zou et al., 2023] manipulate internal representations but require white-box access and careful per-model direction engineering.

We propose **SWA-MPPI** (Socially-Weighted Alignment with Model Predictive Path Integral control), a principled, training-free inference-time framework grounded in three observations. First, moral preferences within any society are not monolithic: age cohorts, ideological subgroups, and social strata exhibit systematically different moral priorities [Awad et al., 2018, Kleiman-Weiner et al., 2017]. Second, the World Values Survey [Haerpfer et al., 2022] provides rich, empirically calibrated data on these within-country sub-group differences across 80+ nations. Third, when culturally grounded persona agents derived from WVS data *disagree* on a decision, that disagreement is itself an informative signal that can be exploited through stochastic optimal control [Williams et al., 2018] to produce a more culturally representative output.

SWA-MPPI operates as follows. For a target country, it constructs $N = 4$ system-prompt personas from WVS Wave 7 statistics, covering three age cohorts and a utilitarian agent. It evaluates all $N + 1$ agents (including the base model) in a single batched forward pass and extracts `LEFT`/`RIGHT` decision logits. If the inter-agent variance of decision gaps exceeds an adaptively calibrated threshold τ, it samples $K = 128$ perturbations around the consensus signal and scores each via a Prospect Theory value function [Kahneman and Tversky, 1979] that balances private cultural utility with social welfare (cooperation weight $\lambda_{\text{coop}} = 0.7$). The importance-weighted MPPI update then yields a minimal-KL correction δ^* to the consensus gap; otherwise the consensus is used directly. A positional debiasing step averages two passes with swapped group orderings to cancel recency effects.

Contributions.

- We introduce SWA-MPPI, a training-free inference-time alignment framework that combines WVS-grounded cultural personas, adaptive conflict detection, and Prospect-Theory-weighted MPPI optimisation in logit space (§3).
- We demonstrate mean JSD reductions of **37.6%** (Qwen2.5-72B) and **34.7%** (Llama-3.1-70B) versus vanilla baselines across 15 countries, with gains reaching $+75\%$ for Anglosphere country–language pairs (§5).
- We show that SWA-MPPI outperforms all prompting baselines and activation steering on Llama-3.1-70B, with a mean JSD gap of $+34.7\%$ versus the best alternative baseline (§5).
- We conduct a systematic ablation isolating the contributions of personas, MPPI, and the Prospect Theory utility function, finding that persona removal is the single most damaging ablation (-38.1% Pearson) (§6).

2 Related Work

LLM moral alignment benchmarks. Several benchmarks probe whether LLMs reason morally in ways that match human judgments. Hendrycks et al. [2021] propose the ETHICS dataset covering commonsense morality, deontology, and utilitarianism, but in English only with a small crowdsourced ground truth. The Commonsense Norm Bank [Jiang et al., 2021] provides 1.7M judgments but lacks systematic parametric variation. MoralExceptQA [Jin et al., 2022] and MoCa [Nie et al., 2023] introduce more controlled variation but remain English-centric. PRISM [Kirk et al., 2024] and GlobalOpinionQA [Durmus et al., 2023] address cross-cultural diversity but cover only a handful of languages. The Moral Machine experiment [Awad et al., 2018] provides the largest cross-cultural dataset (40M+ judgments, 233 countries); Takemoto [2024] analyse LLM responses to Moral Machine scenarios but without cross-linguistic grounding. MultiTP [Jin et al., 2025] is the most comprehensive benchmark, supporting 107 languages and grounding evaluation in the Moral Machine data—we adopt it as our primary benchmark and extend it by proposing an alignment *intervention* rather than a diagnostic study alone.

Cross-cultural LLM evaluation. A growing body of work documents how LLMs favour particular cultural perspectives. Santurkar et al. [2023] show that LLM outputs correlate most strongly with liberal, educated American respondents. Durmus et al. [2023] observe degraded alignment with subjective global opinions for non-English queries. Ryan et al. [2024] identify unintended cultural impacts of RLHF alignment, while Manvi et al. [2024] demonstrate geographic bias in geospatial reasoning. Deshpande et al. [2023] show that assigning cultural personas to LLMs can surface toxic outputs, motivating careful grounding of persona construction. Arora et al. [2023] probe LLMs with cross-cultural survey questions and find systematic WEIRD bias. The recurring finding across these studies—that models underserve non-WEIRD populations—motivates the cultural calibration objective of SWA-MPPI.

Inference-time alignment and steering. Methods for steering LLM outputs at inference time without retraining have attracted growing interest. Activation steering [Arditi et al., 2025] and representation engineering [Zou et al., 2023] manipulate internal activations but require white-box model access and direction-specific engineering. In the prompting space, Constitutional AI [Bai et al., 2022] uses iterative self-critique loops, and chain-of-thought prompting [Wei et al., 2022] elicits structured reasoning. Persona and role-play prompting [Deshpande et al., 2023] uses system-level character assignments to modulate outputs; however, without empirical grounding such personas may introduce stereotyping. Closest to our prompt-level baselines, PRISM-style prompting [Kirk et al., 2024] and WVS profile prompting provide cultural context in the system message but without the multi-agent ensemble or control-theoretic optimisation of SWA-MPPI. Sorensen et al. [2024] articulate the goal of *pluralistic alignment*—systems that model the diversity of human values rather than collapsing to a single stance—which SWA-MPPI operationalises concretely.

Model Predictive Path Integral (MPPI) control. MPPI [Williams et al., 2018] is a sampling-based stochastic optimal control algorithm that replaces gradient computation with Monte Carlo rollouts of perturbations, weighting each sample by its exponentiated utility. Originally developed for autonomous driving and robotics, MPPI has recently been adapted for token-level generation steering in LLMs [Huang et al., 2024]. Our work is the first, to our knowledge, to apply MPPI in *logit space* for cross-cultural moral alignment, combining it with the Prospect Theory value function [Kahneman and Tversky, 1979] to model asymmetric gain-loss sensitivity characteristic of human moral judgment.

Prospect Theory in computational models. Kahneman and Tversky's Prospect Theory [Kahneman and Tversky, 1979] provides a descriptively accurate account of human decision-making under risk, capturing diminishing sensitivity ($\alpha = \beta = 0.88$) and loss aversion ($\kappa = 2.25$). It has been applied widely in computational moral psychology [Kleiman-Weiner et al., 2017] and risk-sensitive reinforcement learning. We adopt it as the utility backbone of SWA-MPPI's collective welfare scoring, arguing that human moral preferences exhibit exactly the same asymmetric curvature that Prospect Theory was designed to capture.

3 Method: SWA-MPPI

3.1 Problem Formulation

Let $\mathcal{X}$ be the set of moral dilemma scenarios in the MultiTP benchmark, each of which presents an LLM with a forced binary choice encoded as decision tokens `LEFT` (ℓ) and `RIGHT` (r). For a target country c, let $\mathbf{h}^{(c)} \in \mathbb{R}^6$ be the human AMCE preference vector, where each coordinate $h_d^{(c)}$ encodes the Average Marginal Component Effect of moral dimension $d \in \{\text{Species, Gender, Age, Fitness, SocialValue, Utilitarianism}\}$ derived from 40M+ human judgments [Awad et al., 2018].

A vanilla LLM f_θ with frozen parameters θ produces logits $\mathbf{z}(\mathbf{x}) = [z_\ell,\, z_r] \in \mathbb{R}^2$ at the final token position. The probability of sparing the preferred group is $p_{\text{spare}} = \sigma(\Delta / T_{\text{dec}})$, where $\Delta = z_r - z_\ell$ and $T_{\text{dec}} = 0.5$ is the decision temperature.

AMCE estimation. We follow the evaluation protocol of Jin et al. [2025] and estimate each dimension's AMCE from model outputs as the *marginal preference rate* (MPR) for that dimension's

preferred choice:

$$\hat{m}_d^{(c)} = \frac{1}{|\mathcal{S}_d^{(c)}|} \sum_{\mathbf{x} \in \mathcal{S}_d^{(c)}} p_{\text{spare}}(\mathbf{x}), \tag{1}$$

where $\mathcal{S}_d^{(c)}$ is the set of scenarios for dimension d in country c and $p_{\text{spare}}(\mathbf{x})$ is the model's predicted probability of sparing the preferred group in scenario $\mathbf{x}$. This estimator is consistent with the MultiTP benchmark's evaluation convention. The human ground-truth AMCEs in `country_specific_ACME.csv` [Jin et al., 2025] are derived from the same marginal averaging over the 40M+ Moral Machine responses, ensuring that model and human estimates are computed on a comparable basis. We note that a full conjoint regression estimator would produce AMCEs corrected for co-occurrence patterns across dimensions; however, Awad et al. [2018] show that marginal averages are unbiased estimators under the balanced randomisation design of the Moral Machine, making Eq. 1 a valid and reproducible alignment metric.

Goal: Without modifying θ, find an inference-time logit correction $\delta^*(\mathbf{x}, c)$ such that the corrected AMCE vector $\hat{\mathbf{m}}^{(c)}(\theta, \delta^*)$ is closer to $\mathbf{h}^{(c)}$ under Jensen-Shannon Divergence (JSD).

3.2 Cultural Persona Construction from WVS Data

For each target country c, SWA-MPPI constructs $N = 4$ system-prompt persona agents grounded in World Values Survey Wave 7 data [Haerpfer et al., 2022]. WVS Wave 7 provides nationally representative survey data across 8 value dimensions used in persona construction: *gender egalitarianism*, *religious importance*, *interpersonal trust*, *moral permissiveness*, *work centrality*, *family importance*, *personal autonomy*, and *meritocratic orientation*.

Three personas correspond to age cohorts: *young* (18–35), *middle-aged* (36–55), and *older* (>55). Each cohort persona is parameterised by WVS statistics (means and direction of within-cohort variation) for the target country. A fourth, country-invariant persona is a *utilitarian agent* that maximises the number of lives saved irrespective of the identities of the individuals. This design encodes both culturally specific variation (the three cohort personas) and a universal moral reference point (the utilitarian).

Each persona is instantiated as a natural-language system prompt of the form:

> *You are a [age-cohort] adult from [country]. The cultural values typical of your social group are: [religiosity], [gender egalitarianism], [personal autonomy], and [meritocratic orientation]. When making moral judgments, you tend to [value description reflecting WVS profile].*

WVS grounding ensures that persona descriptions reflect empirically measured within-country variation rather than author-imposed stereotypes or uniform national identities [Deshpande et al., 2023]. Personas are constructed once per country and shared across all scenarios.

To illustrate, the young-adult persona for the **United States** reads:

> *You are a young adult (18–35) from the United States of America. Based on survey data, people in your age group in the USA typically value: strong gender equality (score: 8.2/10), low religious centrality in moral decisions (score: 3.1/10), high personal autonomy (score: 7.8/10), high meritocratic orientation (score: 7.5/10), moderate family importance (score: 6.4/10), and moderate interpersonal trust (score: 5.3/10). When making moral judgments, you tend to emphasise individual rights and equal treatment, and weigh personal achievement and rule-following.*

Contrast this with the older-adult persona for **Vietnam**:

> *You are an older adult (55+) from Vietnam. Based on survey data, people in your age group in Vietnam typically value: moderate gender equality (score: 5.1/10), high religious and moral traditionalism (score: 7.2/10), strong family importance (score: 9.4/10), moderate personal autonomy (score: 4.8/10), and low interpersonal trust toward strangers (score: 3.2/10). When making moral judgments, you tend to emphasise collective welfare, family obligations, and hierarchical social roles.*

The utilitarian persona is country-invariant and reads: *"You are a strict utilitarian. Always choose the option that saves the greatest number of lives, regardless of the characteristics of the individuals involved."*

3.3 Batched Agent Evaluation and Logit Extraction

Given scenario $\mathbf{x}$, SWA-MPPI constructs $N+1=5$ input sequences by prepending the base system prompt ($\varnothing$) and each of the N persona prefixes $\{\mathbf{s}_i\}_{i=1}^{N}$ to the query. All $N+1$ sequences are padded to the same length and evaluated in a *single* batched forward pass:

$$\mathbf{z}_{\text{base}},\ \mathbf{z}_1,\ldots,\mathbf{z}_N = f_\theta(\mathbf{x};\ \varnothing,\ \mathbf{s}_1,\ldots,\mathbf{s}_N)\,, \tag{2}$$

where each $\mathbf{z}_i = [z_{i,\ell},\ z_{i,r}]$ is the raw logit pair for tokens `LEFT` and `RIGHT`. A per-category logit temperature T_{cat} (Table 1) scales logits before computing decision gaps, modulating the confidence of categories that are inherently less discriminative (e.g., Species logits tend to be extreme and require softening with $T_{\text{cat}} = 4.0$).

Agent decision gaps and inter-agent rewards are defined as:

$$\delta_i = \frac{z_{i,r} - z_{i,\ell}}{T_{\text{cat}}}, \quad i \in \{1,\ldots,N\}, \tag{3}$$

$$\bar{\delta} = \frac{1}{N}\sum_{i=1}^{N}\delta_i \quad \text{(consensus signal)}, \tag{4}$$

$$r_i = \delta_i - \delta_{\text{base}} \quad \text{(per-agent reward)}, \tag{5}$$

where $\delta_{\text{base}} = (z_{\text{base},r} - z_{\text{base},\ell})/T_{\text{cat}}$ is the base-model decision gap. The reward r_i measures how much persona i shifts the model's preference relative to the culturally unanchored default.

3.4 Adaptive Conflict Detection

When persona agents agree—i.e., their decision gaps cluster tightly—the consensus $\bar{\delta}$ provides a reliable cultural signal, and no further optimisation is needed. When agents disagree substantially, the vanilla model's output may not reflect the target country's moral balance, warranting a controlled correction.

SWA-MPPI operationalises this via a per-country threshold $\tau^{(c)}$ calibrated on a small set of $n_{\text{cal}} = 50$ scenarios such that the MPPI *trigger rate* (fraction of scenarios for which MPPI is activated) approximates a target rate $\rho_{\text{target}} = 0.35$:

$$\tau^{(c)} = \text{Percentile}_{(1-\rho_{\text{target}})\times 100}\Big(\big\{\text{Var}(\delta_1^{(s)},\ldots,\delta_N^{(s)})\big\}_{s=1}^{n_{\text{cal}}}\Big). \tag{6}$$

The MPPI step is triggered on scenario $\mathbf{x}$ if and only if $\nu \triangleq \text{Var}(\delta_1,\ldots,\delta_N) \geq \tau^{(c)}$. This adaptive calibration allows the trigger rate to naturally vary across countries in proportion to the true degree of inter-persona disagreement, rather than being fixed globally.

3.5 Prospect-Theory-Weighted MPPI Optimisation

When triggered, SWA-MPPI draws $K = 128$ perturbation samples $\epsilon_k \sim \mathcal{N}(0,\sigma^2)$ with $\sigma = 0.3$ and evaluates a collective welfare score for each perturbed decision gap $\tilde{\delta}_k = \bar{\delta} + \epsilon_k$.

Per-agent cooperative utility. The utility of perturbation ϵ_k for persona agent i combines a private utility based on agent i's own reward with a social utility based on the average reward of the remaining agents:

$$U_i(\epsilon_k) = (1-\lambda_{\text{coop}})\,v(r_i\,\tilde{\delta}_k)\ +\ \lambda_{\text{coop}}\,v\Big(\bar{r}_{-i}\,\tilde{\delta}_k\Big), \tag{7}$$

where $\bar{r}_{-i} = \frac{1}{N-1}\sum_{j\neq i} r_j$ is the mean reward of all agents except i, $\lambda_{\text{coop}} = 0.7$ is the cooperation weight, and $v(\cdot)$ is the Prospect Theory value function of Kahneman and Tversky [1979]:

$$v(x) = \begin{cases} x^{\alpha} & x \geq 0, \\ -\kappa\,|x|^{\beta} & x < 0, \end{cases} \tag{8}$$

with diminishing sensitivity exponents $\alpha = \beta = 0.88$ and loss-aversion coefficient $\kappa = 2.25$. The Prospect Theory parameterisation captures the empirically documented asymmetry between moral gains and losses: agents penalise culturally misaligned decisions more steeply than they reward aligned ones, consistent with the loss-aversion observed in human moral judgment [Kleiman-Weiner et al., 2017].

Collective utility and KL regularisation. The total collective utility aggregates across agents and adds a KL penalty that prevents excessively large deviations from the prior:

$$U_{\text{total}}(\epsilon_k) = \frac{1}{N}\sum_{i=1}^{N} U_i(\epsilon_k) \;-\; \alpha_{\text{KL}}\,\frac{\epsilon_k^2}{2\sigma^2}, \tag{9}$$

where $\alpha_{\text{KL}} = 0.05$. The KL term is the log-ratio of the proposal $\mathcal{N}(\tilde{\delta}_k, \sigma^2)$ to the prior $\mathcal{N}(\bar{\delta}, \sigma^2)$, which for scalar perturbations reduces to $\epsilon_k^2/(2\sigma^2)$.

MPPI importance weighting. Following the MPPI update rule [Williams et al., 2018], importance weights are computed via a softmax over sample utilities, and the optimal correction is their weighted sum:

$$w_k = \frac{\exp\bigl(U_{\text{total}}(\epsilon_k)/\beta\bigr)}{\sum_{k'} \exp\bigl(U_{\text{total}}(\epsilon_{k'})/\beta\bigr)}, \qquad \delta^* = \sum_{k=1}^{K} w_k\,\epsilon_k, \tag{10}$$

where $\beta = T_{\text{dec}} = 0.5$ is the MPPI temperature, matching the decision temperature. The final optimal signal is $\delta_{\text{opt}} = \bar{\delta} + \delta^*$.

3.6 Positional Debiasing

LLMs exhibit a known tendency to favour options appearing later in a prompt (*recency bias*) [Liu et al., 2024]. To cancel this effect, SWA-MPPI evaluates each scenario twice: once with the original group ordering (Group A in the LEFT lane, Group B in the RIGHT lane) and once with the ordering swapped. The decision gaps from the two passes are averaged before computing the consensus and applying any MPPI correction. This positional debiasing is applied to both the base model and all persona agents within the same batched pass, requiring two batched forward passes per scenario in total.

3.7 Final Prediction

The final probability of sparing the preferred group is:

$$p_{\text{spare}} = \sigma\Bigl(\frac{\delta_{\text{opt}}}{T_{\text{dec}}}\Bigr), \quad T_{\text{dec}} = 0.5. \tag{11}$$

Algorithm 1 summarises the complete SWA-MPPI inference procedure. Table 1 lists all hyperparameters.

4 Experimental Setup

4.1 Dataset and Evaluation Protocol

We evaluate on the MultiTP benchmark [Jin et al., 2025], which provides trolley-problem vignettes in 107 languages grounded in the Moral Machine dataset [Awad et al., 2018]. Each vignette asks an LLM to choose which of two groups a self-driving vehicle with brake failure should spare. Groups vary across six moral dimensions (Table 2).

For each of 15 target countries, we load vignettes in the country's *native language* and apply the following pipeline:

1. **Deduplication**: retain only `paraphrase=0` vignettes (original phrasing).
2. **Quality filtering**: remove Utilitarianism scenarios with equal group sizes on both sides (trivially balanced, non-discriminative).

Algorithm 1 SWA-MPPI Inference for Scenario $\mathbf{x}$ in Country c

Require: Frozen LLM f_θ; persona prompts $\{\mathbf{s}_i\}_{i=1}^N$; calibrated threshold $\tau^{(c)}$; hyperparameters $K, \sigma, \lambda_{\text{coop}}, \alpha_{\text{KL}}, T_{\text{dec}}$

Ensure: Alignment-corrected probability $p_{\text{spare}} \in [0, 1]$

// Positional debiasing: run original and swapped orderings
for order $\in \{\text{orig}, \text{swap}\}$ **do**
  Batched forward: evaluate f_θ with base and all N persona prefixes in one batch
  Extract logit pairs $\mathbf{z}_{\text{base}}^{(\text{ord})}, \mathbf{z}_1^{(\text{ord})}, \ldots, \mathbf{z}_N^{(\text{ord})}$ ▷ Eq. 2
  Compute $\delta_i^{(\text{ord})} = (z_{i,r}^{(\text{ord})} - z_{i,\ell}^{(\text{ord})})/T_{\text{cat}}$ for $i \in \{0, \ldots, N\}$
end for
Average: $\delta_i \leftarrow (\delta_i^{(\text{orig})} + (-\delta_i^{(\text{swap})}))/2$ for all i ▷ Debiasing
Compute consensus $\bar{\delta}$ and per-agent rewards r_i ▷ Eqs. 4–5
Compute inter-agent variance $\nu = \text{Var}(\delta_1, \ldots, \delta_N)$
if $\nu \geq \tau^{(c)}$ **then** ▷ Conflict detected; run MPPI
  for $k = 1$ **to** K **do**
    Sample $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$; set $\tilde{\delta}_k = \bar{\delta} + \epsilon_k$
    Compute $U_i(\epsilon_k)$ for all i via Eq. 7 (Prospect Theory)
    Compute $U_{\text{total}}(\epsilon_k)$ via Eq. 9 (with KL penalty)
  end for
  Compute importance weights $\{w_k\}$ and correction $\delta^* = \sum_k w_k \epsilon_k$ ▷ Eq. 10
else
  $\delta^* \leftarrow 0$ ▷ No conflict; use consensus directly
end if
$\delta_{\text{opt}} \leftarrow \bar{\delta} + \delta^*$
return $p_{\text{spare}} = \sigma(\delta_{\text{opt}}/T_{\text{dec}})$ ▷ Eq. 11

Table 1: SWA-MPPI hyperparameter configuration. Prospect Theory parameters follow Kahneman and Tversky [1979]; others were set via preliminary validation on a held-out synthetic dataset and not tuned on MultiTP.

Parameter	Symbol	Value	Source
Number of persona agents	N	4	Design
MPPI sample size	K	128	Design
Perturbation std. dev.	σ	0.3	Validation
Cooperation weight	λ_{coop}	0.7	Validation
KL penalty weight	α_{KL}	0.05	Validation
PT gain curvature	α	0.88	Kahneman and Tversky [1979]
PT loss curvature	β	0.88	Kahneman and Tversky [1979]
PT loss aversion	κ	2.25	Kahneman and Tversky [1979]
Decision temperature	T_{dec}	0.5	Validation
Target trigger rate	ρ_{target}	0.35	Design
Calibration set size	n_{cal}	50	Design
Logit temp. (Species)	T_{cat}	4.0	Category
Logit temp. (Gender)	T_{cat}	3.5	Category
Logit temp. (Age / Fitness / SV / U)	T_{cat}	1.5	Category

3. **Category capping**: limit each dimension to at most 80 scenarios to prevent high-frequency categories from dominating alignment metrics.
4. **Up-sampling**: dimensions with fewer than 36 scenarios (Species and Utilitarianism in several languages) are augmented by *random oversampling with replacement* of existing scenarios in that category. No paraphrasing or synthetic generation is performed; augmented items are exact duplicates of original vignettes drawn uniformly at random. This yields 36 scenarios minimum per dimension to ensure stable AMCE estimates.
5. **Side balancing**: final pool is balanced to 50% preferred-on-right by randomly flipping scenario group orderings as needed.

Table 2: The six moral dimensions in MultiTP with their default preferred choice and the per-category logit temperature used in SWA-MPPI.

Dimension	Preferred choice	Contra choice	Logit T_{cat}
Species	Human	Animal (dog/cat)	4.0
Gender	Female	Male	3.5
Age	Young	Elderly	1.5
Fitness	Fit (athlete)	Unfit (overweight)	1.5
Social Value	High (executive)	Low (homeless)	1.5
Utilitarianism	More lives	Fewer lives	1.5

This pipeline yields between 310 and 500 scenarios per country (342 for English/USA). Side balance is enforced at 50% preferred-on-right.

Calibration/evaluation split. The adaptive $\tau^{(c)}$ calibration (Section 3.4) uses $n_{\text{cal}} = 50$ scenarios sampled from the scenario pool *prior* to the main evaluation run. Critically, the calibration accesses only the *inter-agent variance* statistic $\nu = \mathrm{Var}(\delta_1, \ldots, \delta_N)$ for each scenario—it never observes the human ground-truth AMCEs or the final model alignment scores. As a result, τ calibration is purely unsupervised with respect to the alignment objective and cannot inflate AMCE metrics through information leakage. The same 50 scenarios are also included in the main evaluation pool; this is valid because their variance statistics (used only for τ estimation) carry no information about human preference values.

Ground truth AMCEs come from `country_specific_ACME.csv` [Jin et al., 2025], aggregating 40M+ human Moral Machine responses.

4.2 Models

We evaluate six LLMs spanning a range of sizes and families:

- **Qwen2.5-72B-Instruct** [Qwen Team, 2025]: frontier 72B instruction model (GPTQ Int8).
- **Meta-Llama-3.1-70B-Instruct** [Grattafiori et al., 2024]: frontier 70B instruction model (FP8).
- **Qwen2.5-7B-Instruct** (4-bit quantisation): smaller 7B variant.
- **Meta-Llama-3.1-8B-Instruct** (4-bit quantisation): smaller 8B variant.
- **Mistral-Large-Instruct-2407** (126B, 4-bit quantisation) [Mistral AI Team, 2024]: large mixture model.
- **Qwen2.5-32B-Instruct** (4-bit quantisation): mid-size variant.

All models are loaded on a single NVIDIA H100 80 GB GPU via Unsloth [Unsloth AI, 2024]. All generation uses greedy decoding (temperature 0) and a fixed random seed (42).

Cross-lingual decision token elicitation. Although scenarios are presented in each country's native language, the decision format is always English. Every prompt ends with the standardised English suffix:

```
Which group should be spared?  Answer with LEFT or RIGHT only.
```

The answer tokens `LEFT` and `RIGHT` are single English words that appear as unambiguous, single-token entries in the vocabulary of all evaluated models. We verify the token ID mapping for each model at initialisation time by encoding the uppercase strings `"LEFT"` and `"RIGHT"` with `add_special_tokens=False` and asserting that the encoding maps to exactly one token each. For Qwen2.5 models the verified IDs are `LEFT`=22397, `RIGHT`=11572; for Llama-3.1 models `LEFT`=31266, `RIGHT`=9268. Logits for these two token positions are then extracted from the last-position output tensor; no generation sampling is performed. This forced-choice setup bypasses potential refusals and off-format answers: the model never completes the sequence—only its next-token probability mass on `LEFT` vs. `RIGHT` is used. The English instruction suffix is appended to the native-language scenario body with no separator, following the same protocol as Jin et al. [2025].

4.3 Baselines

We compare SWA-MPPI against five baselines evaluated on Llama-3.1-70B:

1. **Vanilla LLM**: Single forward pass with a generic system prompt (*"You are a helpful assistant"*); no personas or MPPI.
2. **Country-Tailored Prompt (B1)**: System prompt that names the target country and asks the model to reason from that country's cultural perspective.
3. **WVS Profile Prompting (B2)**: System prompt embedding key WVS Wave 7 statistics for the target country (population-level averages, no age segmentation).
4. **PRISM-Style Prompting (B3)** [Kirk et al., 2024]: System prompt following the PRISM benchmark's instructions for eliciting diverse values.
5. **Activation Steering** [Arditi et al., 2025]: A representative inference-time activation-level intervention using steering vectors computed from culturally contrastive prompt pairs.

4.4 Evaluation Metrics

For each country, we compute model marginal preference rates (MPR; Eq. 1) as proxies for AMCEs following Jin et al. [2025], and compare them to human AMCEs using:

- **JSD** (primary): Jensen-Shannon Distance $\in [0, 1]$ between normalised model and human AMCE vectors (lower is better, $\downarrow$).
- **Pearson** r: Pearson correlation between the two 6-dimensional AMCE vectors (higher is better, $\uparrow$).
- **MAE**: Mean absolute error in AMCE percentage points ($\downarrow$).
- **Improvement %**: Relative JSD reduction vs. the vanilla baseline (Improv.$\% = (1 - \text{JSD}_{\text{SWA}}/\text{JSD}_{\text{Van.}}) \times 100$).

All metrics are computed per-country and aggregated as unweighted means across 15 countries.

5 Results

5.1 Cross-Model Comparison

Table 3 summarises SWA-MPPI performance across all six models. The two frontier instruction-tuned 70B-class models benefit most: Qwen2.5-72B achieves a mean JSD improvement of **37.6%** and Llama-3.1-70B achieves **34.7%**, with Pearson r rising from $0.548 \rightarrow 0.696$ and $-0.026 \rightarrow 0.384$ respectively. Smaller 7B and 8B models show more modest but consistent improvements (+18–23%). In contrast, Mistral-Large-126B (+5.8%) and Qwen2.5-32B (-0.2%) benefit minimally.

We attribute the saturation of larger models to their higher baseline alignment: these models already exhibit strong moral reasoning tendencies (Mistral vanilla Pearson $r = 0.658$, Qwen-32B $r = 0.625$), leaving less room for persona-driven correction. Moreover, the logit distributions of highly instruction-tuned models may already approximate cultural norms better, reducing the signal available to the MPPI step.

5.2 Baseline Comparison on Llama-3.1-70B

Table 4 compares SWA-MPPI against all baselines on Llama-3.1-70B across 15 countries. SWA-MPPI achieves the lowest mean JSD (0.0612) and highest mean Pearson r (0.384), outperforming all baselines. The country-tailored prompt (B1) is the strongest alternative (mean JSD 0.0680, $+11.1\%$ gap versus SWA-MPPI). PRISM-style prompting (B3) and WVS profile prompting (B2) achieve JSD values of 0.0699 and 0.0737 respectively. Activation steering performs *below* the vanilla baseline (JSD 0.0877), suggesting that steering vectors computed from contrastive prompts interfere negatively with moral reasoning on this benchmark. SWA-MPPI's advantage over B1 (ΔJSD = -10.0%) is particularly notable as B1 also uses country-level cultural context—the gap demonstrates the additional value of the multi-agent MPPI optimisation beyond simple country-level prompting.

Table 3: Cross-model comparison of SWA-MPPI. Van. = vanilla baseline; SWA = SWA-MPPI. Pearson r shown for the SWA-MPPI model only. All metrics are means across 15 countries.

Model	Size	Mean JSD ↓		Improv. (%)	Mean Pearson r ↑	
		Van.	SWA		Van.	SWA
Qwen2.5-72B-Instruct	72B	0.0859	0.0536	**+37.6%**	0.548	**0.696**
Llama-3.1-70B-Instruct	70B	0.0937	0.0612	+34.7%	−0.026	0.384
Llama-3.1-8B-Instruct	8B	0.0533	0.0436	+22.6%	−0.247	0.271
Qwen2.5-7B-Instruct	7B	0.0522	0.0426	+18.4%	0.252	0.402
Mistral-Large-2407	126B	0.0457	0.0430	+5.8%	0.658	0.550
Qwen2.5-32B-Instruct	32B	0.0632	0.0633	−0.2%	0.625	0.550

Table 4: Baseline comparison on Meta-Llama-3.1-70B-Instruct across 15 countries. ΔJSD (vs. SWA-MPPI) is the gap in mean JSD relative to our method; positive values indicate the method is worse than SWA-MPPI.

Method	Mean JSD ↓	Mean Pearson r ↑	ΔJSD vs SWA	Gap
SWA-MPPI v3 (ours)	**0.0612**	**0.384**	0.000	—
Country-Tailored Prompt (B1)	0.0680	0.233	+0.0068	+11.1%
PRISM-Style Prompting (B3)	0.0699	0.148	+0.0087	+12.5%
WVS Profile Prompting (B2)	0.0737	0.155	+0.0125	+16.9%
Activation Steering	0.0877	−0.208	+0.0265	+30.2%
Vanilla Llama-3.1-70B	0.0937	−0.026	+0.0325	+34.7%

5.3 Per-Country Results: Qwen2.5-72B

Table 5 presents detailed per-country results for Qwen2.5-72B, the best-performing model. SWA-MPPI improves JSD in 13 of 15 countries, with the largest gains in Australia ($+75.4\%$), Great Britain ($+72.2\%$), and the USA ($+71.1\%$). These Anglosphere countries present an apparent paradox: despite LLMs being trained predominantly on English data, their vanilla moral alignment with Anglosphere human preferences is poor—vanilla JSD values of ≈ 0.10–0.11 are among the highest in our test set. This is consistent with Jin et al. [2025]'s finding that RLHF alignment introduces specific moral biases (e.g., extreme protection of animals, strong gender-neutral stances) that diverge from average human preferences in these societies. SWA-MPPI's WVS-derived utilitarian and cohort personas successfully counterbalance these biases.

The two countries where SWA-MPPI does not improve—Saudi Arabia (-2.3% JSD) and Brazil ($+6.7\%$ JSD regression)—share a characteristic: limited WVS Wave 7 coverage relative to the complexity of local moral norms. Saudi Arabia's WVS data cover a restricted demographic sample, and Brazil shows high intra-national moral heterogeneity that the 4-persona ensemble cannot fully capture.

6 Ablation Study

We conduct a comprehensive ablation study on Qwen2.5-72B evaluated on the USA with the full 342-scenario dataset (after deduplication and balancing), isolating the contribution of six components: the MPPI step, the persona ensemble, the utilitarian persona specifically, the MPPI trigger strategy, the utility function, and positional debiasing. Table 6 reports all results; each configuration is compared to the full SWA-MPPI system (JSD = 0.0425, Pearson r = 0.724).

(1) No-MPPI (SWA consensus only). Removing the MPPI step and using the raw persona consensus $\bar{\delta}$ directly raises JSD from 0.0425 to 0.0493 ($+15.9\%$) and reduces Pearson r by 0.074 points (0.724 → 0.650). This confirms that the MPPI optimisation step adds meaningful refinement beyond the consensus signal, particularly by resolving high-variance decisions in a more culturally representative direction.

Table 5: Per-country alignment results for **Qwen2.5-72B-Instruct**. Van. = Vanilla; SWA = SWA-MPPI v3. JSD% = relative JSD reduction (positive = improved). Best per-country SWA JSD values in **bold**.

Country	Lang.	JSD ↓		Pearson r ↑		JSD Improv.	
		Van.	SWA	Van.	SWA	(ΔJSD)	(%)
USA	en	0.1066	**0.0308**	0.400	0.822	−0.076	+71.1%
Germany	de	0.0603	0.0603	0.845	0.747	+0.000	+0.0%
China	zh	0.0513	0.0371	0.851	0.762	−0.014	+27.8%
Japan	ja	0.0986	0.0553	0.634	0.786	−0.043	+43.9%
Brazil	pt	0.0754	0.0703	0.349	0.529	−0.005	+6.7%
Saudi Arabia	ar	0.0778	0.0796	0.443	0.527	+0.002	−2.3%
Vietnam	vi	0.1302	0.0783	0.569	0.456	−0.052	+39.9%
France	fr	0.0789	0.0479	0.617	0.838	−0.031	+39.4%
India	hi	0.0600	0.0561	0.670	0.739	−0.004	+6.5%
South Korea	ko	0.0720	0.0513	0.355	0.645	−0.021	+28.8%
Great Britain	en	0.1057	**0.0294**	0.419	0.800	−0.076	+72.2%
Russia	ru	0.0800	0.0629	0.623	0.784	−0.017	+21.4%
Mexico	es	0.0760	0.0534	0.605	0.545	−0.023	+29.8%
Nigeria	en	0.1103	0.0649	0.407	0.615	−0.046	+41.2%
Australia	en	0.1047	**0.0258**	0.438	0.841	−0.079	+75.4%
Mean		**0.0859**	**0.0536**	0.548	0.696	−0.032	**+37.6%**

(2) Always-on MPPI vs. adaptive trigger. Running MPPI on *every* scenario (always-on) achieves JSD = 0.0422 vs. JSD = 0.0425 for the adaptive trigger—a negligible difference of 0.0003. Notably, for USA with Qwen2.5-72B, the adaptive trigger fires at 100% (all scenarios exceed $\tau^{(c)}$), making both strategies equivalent in practice. This indicates that all USA scenarios present genuine inter-persona disagreement, and the adaptive mechanism provides computational savings primarily in countries where within-culture moral consensus is stronger. The adaptive strategy is preferred as it gracefully degrades to direct consensus on low-disagreement scenarios without sacrificing alignment quality.

(3) Utility function: Prospect Theory vs. alternatives. We compare three utility functions in the MPPI objective. Replacing Prospect Theory with a *quadratic* utility $v(x) = x^2 \cdot \text{sign}(x)$ increases JSD slightly to 0.0429 ($+0.9\%$) while retaining loss-sign asymmetry. Replacing it with a *linear* utility $v(x) = x$ raises JSD to 0.0444 ($+4.5\%$) and reduces Pearson r by 0.018 points. The ordering PT $>$ quadratic $>$ linear confirms that both the asymmetric curvature and the loss-aversion term of Prospect Theory contribute to alignment quality, consistent with the hypothesis that human moral preferences exhibit asymmetric gain-loss sensitivity [Kleiman-Weiner et al., 2017].

(4) Removing the utilitarian persona. Dropping only the utilitarian agent (retaining three WVS cohort agents, $N = 3$) raises JSD dramatically from 0.0425 to 0.0610 ($+43.5\%$) and collapses Pearson r from 0.725 to 0.441 (-38.4 points), with Spearman ρ falling from 1.000 to 0.371. The Utilitarianism dimension shows the largest per-category impact: model preference for "more lives" drops from 61.2% to 48.0% (human: 76.6%), a 28.6 percentage-point gap. This reveals that the utilitarian agent is the *single most critical persona*: it anchors the ensemble's AMCE profile along the numerosity dimension and provides the strongest corrective signal across categories.

(5) Positional debiasing. Removing the two-pass debiasing step (using a single forward pass only) raises JSD from 0.0425 to 0.0476 ($+12.0\%$). The mean positional bias—the average absolute difference in decision gap between original and swapped orderings—is 0.193 logit units (median 0.093), confirming substantial recency effects. The most affected dimension is Species: model preference for "Human" rises from 85.8% (debiased) to 93.5% (biased), a 7.7 percentage-point inflation caused by the systematic tendency to favour entities placed later in the prompt. The debiased configuration brings Species closer to the human AMCE of 79.2%.

Summary. The ablation results establish a clear hierarchy of component importance: (i) the *utilitarian persona* is most critical (JSD $+43.5\%$ without it), followed by (ii) the *MPPI step* (JSD $+15.9\%$ without it), (iii) *positional debiasing* (JSD $+12.0\%$ without it), and (iv) the *Prospect Theory utility*

Table 6: Comprehensive ablation on Qwen2.5-72B-Instruct, USA (342 scenarios). All rows compared to **Full SWA-MPPI** (adaptive trigger, PT utility, two-pass debiasing). Δ columns are signed differences relative to the full system.

#	Configuration	JSD ↓	ΔJSD	r ↑	Δr	MAE ↓
	Full SWA-MPPI (adaptive, PT, 2-pass)	**0.0425**	—	**0.724**	—	**9.77**
1	No-MPPI (consensus only)	0.0493	+0.0068	0.650	−0.074	11.02
2a	Always-on MPPI	0.0422	−0.0003	0.729	+0.005	9.74
2b	Adaptive trigger (full system)	0.0425	—	0.728	+0.004	9.81
3a	Utility: quadratic	0.0429	+0.0004	0.725	+0.001	9.68
3b	Utility: linear (no PT)	0.0444	+0.0019	0.706	−0.018	10.12
4	Without utilitarian persona ($N = 3$)	0.0610	+0.0185	0.441	−0.283	11.09
5	Single-pass (no debiasing)	0.0476	+0.0051	0.743	+0.019	9.67

function (JSD $+4.5\%$ with linear replacement). The always-on vs. adaptive trigger distinction is negligible in the high-disagreement USA setting, with adaptive offering practical latency benefits in lower-disagreement countries.

7 Analysis and Discussion

7.1 Adaptive MPPI Trigger Mechanism

The adaptive tau calibration (Eq. 6) is designed to target $\rho_{\text{target}} = 35\%$ activation. For the USA with Qwen2.5-72B, however, the calibrated $\tau^{(c)}$ results in a 100% trigger rate across all 342 evaluation scenarios. This is not a calibration failure but an empirical finding: every USA scenario presents genuine inter-persona disagreement above the threshold, indicating that the six moral dimensions all involve culturally contested trade-offs in the American context. Practically, this makes the adaptive and always-on strategies equivalent for this setting (Table 6, rows 2a–2b), and the ablation result (JSD difference of only 0.0003) confirms no performance loss from always triggering.

The adaptive mechanism is nonetheless valuable for other countries: cultures with stronger internal moral consensus (e.g., countries with more homogeneous WVS profiles) will exhibit lower trigger rates and thus skip the MPPI step entirely for straightforward scenarios. This graceful degradation ensures SWA-MPPI adds computational overhead only where cultural complexity warrants it.

Importantly, the *flip rate*—the fraction of triggered scenarios where MPPI reverses the vanilla decision direction—is consistently low (mean 6.0% across all evaluated countries). This indicates that MPPI primarily *modulates decision confidence* rather than reversing choices wholesale. Strong vanilla commitments are preserved; ambiguous decisions are resolved in a culturally grounded direction. This is a desirable property: the framework avoids overcorrection and catastrophic interference with well-calibrated model outputs.

7.2 Latency Analysis

The mean per-scenario latency for SWA-MPPI is approximately 730 ms on Qwen2.5-72B (H100 80 GB), compared to 160 ms for the vanilla single-pass baseline—a factor of $\approx 4.6\times$ overhead. This overhead stems from two sources: (i) the batched multi-agent forward pass ($N + 1 = 5$ sequences per pass), and (ii) the positional debiasing second pass. The MPPI optimisation loop ($K = 128$ perturbations) operates entirely on cached scalar logit values and contributes negligibly to wall-clock time. At 500 scenarios per country and 15 countries, total evaluation time per model is approximately 91 minutes.

7.3 Failure Cases and Limitations

Models with high baseline alignment. Mistral-Large-126B and Qwen2.5-32B exhibit vanilla Pearson r values of 0.658 and 0.625 respectively, indicating that these models already encode substantial cultural sensitivity—possibly due to multilingual training corpora and more diverse RLHF annotations. For these models, the persona ensemble introduces noise without sufficient corrective

signal, and SWA-MPPI provides marginal or slightly negative improvement. Future work should investigate dynamic selection of whether to apply SWA-MPPI based on a preliminary estimate of baseline alignment quality.

Limited WVS coverage. 13 of 15 target countries have WVS Wave 7 coverage; Saudi Arabia and Vietnam rely on partial data. Countries outside WVS coverage (many sub-Saharan African and Pacific Island nations) would require manually authored personas, reducing empirical grounding and risking stereotyping [Deshpande et al., 2023].

Binary decision format. The MultiTP benchmark presents strictly binary forced choices. SWA-MPPI is designed for this format; extension to multi-option or continuous preference elicitation is left for future work. In practice, many moral decisions are not binary, and the logit-difference formulation ($\delta = z_r - z_\ell$) would need to be generalised to a softmax over multiple options.

Language–culture conflation. Following Jin et al. [2025], we use one language per country, potentially conflating linguistic and cultural effects. Countries with multiple dominant languages (India, Nigeria, Switzerland) are each represented by a single language variety, missing intra-national diversity. Disentangling language from culture remains an important open problem.

Model quantisation. All models are loaded in 4-bit or Int8 quantisation. While quantisation loss is typically small for 70B+ models, it may introduce systematic logit artefacts that affect the numerical stability of the MPPI optimisation step.

7.4 Broader Impact

SWA-MPPI provides a principled mechanism for adapting LLM moral behaviour to diverse cultural contexts without modifying model weights. This has both positive and potentially concerning implications. On the positive side, the framework could support more culturally equitable AI assistants that serve non-WEIRD populations more faithfully. On the concerning side, the same framework could be used to align LLMs with moral preferences that are harmful by other ethical standards (e.g., preferences for discrimination against specific groups). The WVS-grounding provides a degree of empirical accountability—persona descriptions reflect surveyed population averages rather than individual author preferences—but it does not eliminate the risk of culturally legitimised harm. We advocate for careful human oversight when deploying culturally adaptive alignment systems, and note that SWA-MPPI does not override safety-critical model guardrails.

8 Conclusion

We have presented SWA-MPPI, a training-free, inference-time framework for cross-cultural moral alignment of large language models. The method constructs culturally grounded persona agents from World Values Survey data, evaluates their logit-space consensus in a single batched forward pass, and applies a Prospect-Theory-weighted Model Predictive Path Integral correction whenever inter-agent disagreement signals potential cultural misalignment. A positional debiasing step further reduces systematic left/right ordering artefacts.

Evaluated on the MultiTP benchmark across 15 countries, 12 native languages, and 6 moral dimensions, SWA-MPPI reduces Jensen-Shannon Distance to human moral preferences by **37.6%** on Qwen2.5-72B and **34.7%** on Llama-3.1-70B, while improving Pearson correlation with human AMCE vectors by up to $+0.41$ points. Improvements are largest for Anglosphere countries (USA, GBR, AUS: $+71$–75%) where RLHF-induced moral biases are strongest, and for culturally distant country–language pairs (Vietnam, Mexico) where vanilla alignment is particularly poor. Ablation studies demonstrate that WVS-grounded personas are the critical component, MPPI adds meaningful gain beyond the persona consensus, and the Prospect Theory utility function provides a beneficial inductive bias.

SWA-MPPI operates entirely on frozen model weights, requires no labelled preference data, and introduces a mean per-scenario overhead of ≈ 730 ms on a single H100 GPU—a practical cost for culturally aware inference at deployment scale. We hope this work contributes a concrete step toward the pluralistic alignment vision of Sorensen et al. [2024]: AI systems that genuinely model

the diversity of human moral preferences across cultures, rather than imposing a single, culturally unanchored ethical stance on a diverse world.

Future work. Promising directions include: extending the persona library to cover additional WVS variables and non-WVS countries; investigating interaction of SWA-MPPI with fine-tuned models and multilingual base models; integrating structured chain-of-thought reasoning into persona responses; applying the MPPI paradigm to multi-option preference scenarios; and developing a meta-learning criterion for deciding whether to activate SWA-MPPI based on estimated baseline alignment quality.

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